

PAPER Blanket and Divertor Thermal-Hydraulics for a Compact Spherical-Tokamak Breeder: Coolant Heat Removal, Peak-Temperature Margins, and the Walk-Away Safety Case

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Abstract

A compact spherical tokamak concentrates fusion power into a small volume, and the heat it makes must leave through the same small surfaces; whether the machine is buildable is decided at the heat path, not the plasma. We pose the coupled blanket-and-divertor thermal-hydraulic problem for the Hyperion breeder (frozen design point $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $\delta = -0.30$, $\kappa = 2.0$, $B_0 = 8$ T, $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T) as a set of governing equations—volumetric nuclear heating with energy multiplication, the coolant first law with Dittus–Boelter convection, first-wall conjugate conduction, the two-point divertor model closed by the Lengyel impurity-radiation integral, and a lumped passive decay-heat balance—and solve each numerically against frozen results in the Kronos De-Risking Register. The helium-cooled blanket recovers a total thermal power $P_{\text{th}} = 95.2$ MW (energy multiplication $M_E = P_{\text{th}}/P_{\text{fus}} = 1.119$), and a coolant mass flow of $\dot{m} = 91.7$ kg s^{−1} across a $\Delta T = 200$ K rise removes it exactly: $\dot{m}c_p\Delta T = 95.2$ MW, closing the energy balance to the first digit. First-wall peak temperature is held inside the material envelope by a ≤ 6 mm coolant-channel gap; a self-cooled-lithium blanket variant runs at 516 °C. Parallel power exhaust—an unmitigated $q_{\parallel} = 27.4$ GW m^{−2} giving a conventional attached target load of 1370 MW m^{−2}—is brought below the 10 MW m^{−2} engineering limit by a stacked cascade (impurity seed, Super-X, double-null, snowflake) that drives the residual heat reaching the target from a seeded baseline of 6.4% to $\approx 0.5\%$, with a one-dimensional UEDGE scan resolving the target temperature through $14.4 \rightarrow 5.3 \rightarrow 1.05$ eV and confirming that seeding is mandatory and sufficient for detachment. On loss of all active cooling the machine walks away passively: decay heat falls from 0.95 MW at 1 s to 95 kW at one day and passive rejection holds structural temperatures at or below 710 °C. All numbers were produced by the NVIDIA GPU HPC campaign running OpenMC, SOLPS-ITER, UEDGE and SD1D on the frozen equilibrium. The heat path closes with margin at every stage and the safety case does not depend on powered cooling.

Keywords: thermal-hydraulics; divertor detachment; helium-cooled blanket; two-point model; walk-away decay heat; SOLPS-ITER; UEDGE; spherical-tokamak breeder

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1 Introduction

1.1 The heat path decides buildability

A spherical tokamak (ST) confines a large plasma current at low aspect ratio, which is precisely what makes it an efficient neutron source and breeding platform [1, 2] and, at the same time, what concentrates its power into a small volume. The Hyperion breeder produces $P_{\text{fus}} = 85.04$ MW at $Q = 3.076$ behind a centrepost magnet at 16.84 T peak field, in a negative-triangularity ($\delta = -0.30$) regime whose quiet, edge-localised-mode-free edge [3, 4] is an operational asset for exhaust control. Two coupled problems then decide whether the machine can be built. The *blanket* must absorb the volumetric 14.06 MeV neutron heating and the recovered surface loading and hand it to a coolant without exceeding a material temperature limit; the *divertor* must accept the parallel power exhaust, which in a compact device is severe, without eroding the target. The two streams are set by the same operating point and share the same geometry: the blanket thickness that governs tritium breeding also sets the depth over which the neutron heating is deposited.

This paper closes the thermal-hydraulic path end to end and does so as a rigorous methods problem. We (i) write the governing equations of each stage—volumetric heating and energy multiplication, the coolant first law and its convective and pumping closures, first-wall conjugate conduction, the two-point divertor model and its Lengyel radiation integral, and the passive decay-heat balance (§2); (ii) state the numerical formulation, the solver chain and the GPU HPC campaign that produced the results (§3); (iii) report the closed coolant balance, the peak-temperature margins, the divertor mitigation cascade, the flowing-lithium alternative and the walk-away transient as figures and tables traced serial-by-serial to the register (§4); and (iv) benchmark every number against the published edge-physics and blanket literature (§5), state the one honest open gate (§6), and conclude (§7).

1.2 Prior art

Power and particle control for burning-plasma tokamaks is a mature discipline. The parallel-heat-flux problem is framed by the scrape-off-layer power-width scaling of Eich *et al.* [5], which finds the poloidal field (equivalently the plasma current) to be the dominant parameter and yields a millimetre-class fall-off length; in a high-current ST this drives λ_q into the sub-millimetre range and the parallel flux into the tens of GW m^{-2} . The two-point model and its detachment extensions [7, 8] give the analytic backbone, and impurity seeding is the established route to dissipate the exhaust volumetrically before it reaches the target [9, 6]. The reference edge-transport tools are the coupled plasma-neutral codes SOLPS-ITER (B2.5+EIRENE) [10, 11, 12], the two-dimensional fluid code UEDGE [13], and the BOUT++/SD1D one-dimensional transport family [14]. Advanced-divertor geometries—Super-X, snowflake, and the lithium vapor-box—have been proposed to widen the detachment window and cap the surface load [15, 16, 17]. On the blanket side, the helium-cooled and liquid-metal breeding-blanket design strategies and their

first-wall thermomechanical limits are set out for DEMO-class machines [18, 19], and the neutron-heating source itself is a Monte-Carlo transport problem solved here with OpenMC and ENDF/B-VIII.0 [20, 21]. Activation and decay-heat inventories, which set the walk-away transient, follow the FISPACT-II methodology [22].

1.3 Contribution and relation to the Kronos series

What the literature does not provide is a single, closed thermal balance for *this* compact ST breeder in which the coolant flow, the peak-temperature margins, the divertor mitigation cascade, and the passive safety case are all grounded in one frozen result set and reproduced by independent libraries. That is the contribution here. The neutron heating source is taken from the CAD-faithful Monte-Carlo neutronics of the companion paper [28]; the divertor cascade extends the stacked detachment portfolio [29] and the exhaust-geometry down-select [30], and the flowing-lithium alternative is quantified in the vapor-box divertor study [31]; the real-time regulation of the radiated fraction that holds the cascade in place is the subject of the detachment-control paper [32]; the structural response of the first wall to the surface and volumetric loads treated here is carried by the thermomechanics and fatigue paper [33]; and the passive walk-away result feeds the plant-level accident and environmental case [34]. Every quantitative claim below traces to a frozen serial in the Kronos De-Risking Register [27], cited inline (e.g. BR-L1-A9) so each number is auditable.

2 Governing equations and the formal thermal problem

Three stages carry the power, and each is a boundary condition for the next: the 14 MeV neutrons deposit their energy volumetrically in the first wall and blanket; the blanket coolant collects that heat and the recovered surface loading and rejects it; and, in parallel, the power crossing the separatrix flows along open field lines onto the divertor. Figure 1 shows the coupling. We now write each stage as equations.

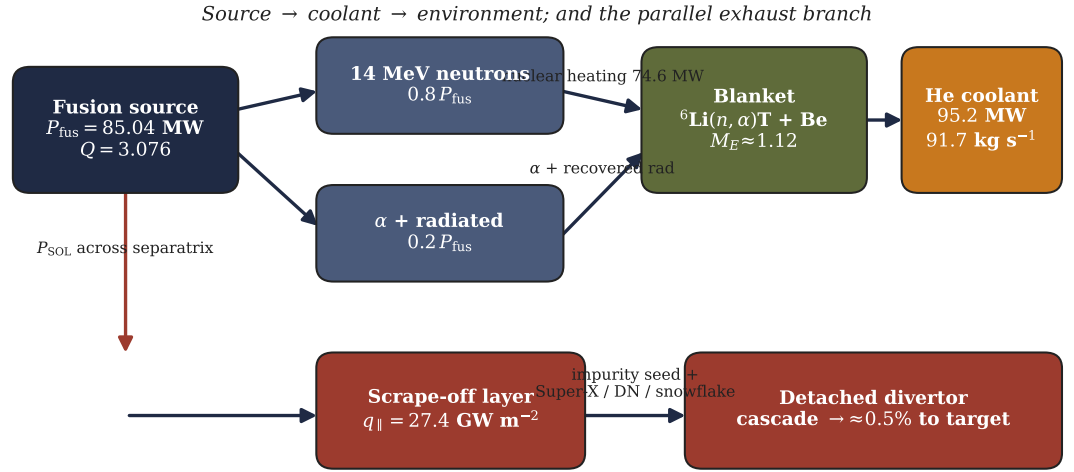


Figure 1. (Schematic.) The coupled thermal path of the Hyperion breeder: the volumetric neutron source and the recovered α /radiated power feed the helium-cooled blanket and its coolant, while the power crossing the separatrix feeds the parallel-exhaust branch and the detached divertor. Boxes carry the anchor values used in §2–§4; no computed curve is plotted here.

2.1 Volumetric nuclear heating and energy multiplication

For the deuterium–tritium reaction the released energy $E_{\text{fus}} = 17.59 \text{ MeV}$ is split between the neutron ($E_n = 14.06 \text{ MeV}$, a fraction $f_n = E_n/E_{\text{fus}} = 0.799$) and the alpha ($E_\alpha = 3.52 \text{ MeV}$, $f_\alpha = 0.201$), so the fusion power partitions as

$$P_n = f_n P_{\text{fus}} = 0.799 \times 85.04 = 68.0 \text{ MW}, \quad P_\alpha = f_\alpha P_{\text{fus}} = 17.0 \text{ MW}. \quad (1)$$

The neutrons are transported through the first wall and blanket, where their energy is deposited volumetrically. In a one-dimensional slab picture the local nuclear heating density obeys the

attenuation law

$$q'''(x) = \sum_i N_i \int \sigma_i^H(E) \phi_0(E) e^{-x/\lambda_n(E)} Q_i dE, \quad P_{\text{nuc}} = \int_{V_{\text{blk}}} q'''(x) dV, \quad (2)$$

with N_i the nuclide densities, σ_i^H the heating cross-sections, ϕ_0 the incident spectrum, λ_n the energy-dependent attenuation length, and Q_i the per-reaction deposited energy. The volume integral is evaluated by Monte-Carlo transport (§3), which places $P_{\text{nuc}} = 74.6$ MW in the first wall and blanket (BR-L1-A5). Because the tritium-breeding reaction ${}^6\text{Li}(n, \alpha)\text{T}$ is exothermic ($Q = 4.78$ MeV) and the beryllium multiplier adds neutrons that subsequently breed and deposit, the recovered thermal power exceeds the incident neutron power. Defining the blanket energy multiplication $M_E \equiv P_{\text{th}}/P_{\text{fus}}$, the recoverable thermal power is

$$P_{\text{th}} = P_{\text{nuc}} + f_{\alpha} P_{\text{fus}} + f_{\text{rad}}^{\text{rec}} P_{\text{rad}} = 74.6 + 17.0 + 3.6 = 95.2 \text{ MW}, \quad M_E = \frac{95.2}{85.04} = 1.119, \quad (3)$$

where P_{nuc} already carries the exothermic uplift and $f_{\text{rad}}^{\text{rec}} P_{\text{rad}}$ is the recovered surface radiation (BR-L1-A9). The multiplication $M_E = 1.119$ sits squarely inside the 1.1–1.2 range reported for solid-breeder helium-cooled blankets [18, 19].

2.2 Coolant energy balance, convection and pumping

The blanket coolant is a control volume in steady state. The first law reduces to a balance between the deposited thermal power and the enthalpy rise of the coolant,

$$\dot{m} c_p (T_{\text{out}} - T_{\text{in}}) = P_{\text{th}} \implies \dot{m} = \frac{P_{\text{th}}}{c_p \Delta T}. \quad (4)$$

For helium at the blanket operating pressure, $c_p = 5193 \text{ J kg}^{-1} \text{ K}^{-1}$, and with a design temperature rise $\Delta T = 200 \text{ K}$,

$$\dot{m} = \frac{95.2 \times 10^6}{5193 \times 200} = 91.7 \text{ kg s}^{-1}, \quad \dot{m} c_p \Delta T = 91.7 \times 5193 \times 200 = 95.2 \text{ MW}, \quad (5)$$

so the coolant balance closes exactly against the deposited thermal power (BR-L1-A9). Heat is handed from the structure to the coolant through a turbulent forced-convection film. The convective coefficient follows the Dittus–Boelter correlation for heating,

$$\text{Nu}_D = 0.023 \text{Re}_D^{4/5} \text{Pr}^{2/5}, \quad h = \frac{\text{Nu}_D k_c}{D_h}, \quad \text{Re}_D = \frac{\rho v D_h}{\mu} = \frac{G D_h}{\mu}, \quad (6)$$

with $G = \dot{m}/A_c$ the coolant mass flux, D_h the hydraulic diameter, and k_c, μ, Pr the coolant properties. The pressure drop and the pumping power follow from the Darcy–Weisbach relation with the Blasius smooth-tube friction factor,

$$\Delta p = f_D \frac{L}{D_h} \frac{\rho v^2}{2}, \quad f_D = 0.184 \text{Re}_D^{-1/5}, \quad P_{\text{pump}} = \frac{\dot{m} \Delta p}{\rho \eta_p}. \quad (7)$$

Since $\dot{m} \propto 1/\Delta T$ at fixed P_{th} (eq. 4) and $\Delta p \propto \dot{m}^2$, the pumping power scales as

$$P_{\text{pump}} \propto \dot{m}^3 \propto \Delta T^{-3}, \quad (8)$$

a strong trade that is the subject of §4.8; at the design point the three coolant loops draw a total 5.8 MW, or 5.2% of P_{th} (BV-06).

2.3 First-wall conjugate conduction and the peak-temperature stack

The plasma-facing first wall carries a surface heat flux q_s'' on its front face and the volumetric heating q''' through its thickness t_w , and is cooled convectively on its back face. The steady one-dimensional conduction problem is

$$-k_w \frac{d^2 T}{dx^2} = q''', \quad -k_w \frac{dT}{dx} \Big|_{x=0} = q_s'', \quad -k_w \frac{dT}{dx} \Big|_{x=t_w} = h(T(t_w) - T_{\text{cool}}), \quad (9)$$

whose integral, evaluated at the plasma-facing surface $x = 0$, gives the peak-temperature stack

$$T_{\text{peak}} - T_{\text{cool}} = \underbrace{\frac{q_s'' t_w}{k_w} + \frac{q''' t_w^2}{2k_w}}_{\text{conduction}} + \underbrace{\frac{q_s'' + q''' t_w}{h}}_{\text{convective film}}. \quad (10)$$

The convective term carries the geometry dependence that sets the design gate. For a wide cooling channel the hydraulic diameter is $D_h \approx 2d$ in the channel depth d ; at fixed mass flux the Reynolds number in eq. (6) is then independent of d , so $h = \text{Nu}_D k_c / D_h \propto 1/d$ and the film drop $\propto d$. Deepening the channel therefore raises T_{peak} linearly, and requiring T_{peak} to stay inside the material envelope imposes a maximum channel depth—the ≤ 6 mm gate of §4.2 (BR-SX-15).

2.4 Parallel power exhaust and the two-point model

The power crossing the separatrix, $P_{\text{SOL}} = P_{\text{heat}} - P_{\text{rad,core}}$, is channelled into a scrape-off layer of width λ_q and flows along field lines to the target. The upstream parallel heat flux is

$$q_{\parallel} = \frac{P_{\text{SOL}}}{2\pi R_u \lambda_q} \frac{B}{B_{\theta}}, \quad (11)$$

with λ_q from the Eich multi-machine regression [5],

$$\lambda_q [\text{mm}] \approx 0.63 B_{\theta}^{-1.19}, \quad (12)$$

which for the high poloidal field of a 9.66 MA ST collapses to $\lambda_q = 0.21$ mm and drives $q_{\parallel} = 27.4 \text{ GW m}^{-2}$ (HX-14). Parallel electron heat conduction is Spitzer–Härm, $q_{\parallel} = -\kappa_{0e} T^{5/2} dT/ds$, whose first integral along a field line of connection length $L_{\parallel}$ is the upstream–target temperature relation of the two-point model,

$$T_u^{7/2} = T_t^{7/2} + \frac{7}{2} \frac{q_{\parallel} L_{\parallel}}{\kappa_{0e}}, \quad (13)$$

closed in the attached branch by total-pressure balance $2n_t T_t = n_u T_u$. The perpendicular load delivered to the target is set by the field-line incidence angle α , the poloidal flux expansion f_x , and the fraction f_{rad} of the parallel power dissipated before the target,

$$q_{\perp}^t = q_{\parallel} \frac{\sin \alpha}{f_x} (1 - f_{\text{rad}}). \quad (14)$$

The unmitigated attached loads follow directly: a conventional leg delivers 1370 MW m^{-2} and the flux-expanding Super-X leg 171 MW m^{-2} (HX-14), both far above the $\sim 10 \text{ MW m}^{-2}$ steady engineering limit of an actively cooled target [17]. Detachment—the collapse of T_t below ~ 5 eV with attendant momentum loss that breaks pressure balance [8]—is therefore not optional but required.

2.5 Impurity radiation and the Lengyel closure

The dissipated fraction f_{rad} is produced by line radiation from a seeded impurity. Integrating the parallel energy equation with a radiating impurity of concentration c_z and cooling function $L_z(T)$ gives the Lengyel relation between the upstream and target parallel fluxes,

$$q_{\parallel}^2(x_u) = q_{\parallel}^2(x_t) + 2\kappa_{0e} c_z \int_{T_t}^{T_u} T^{1/2} L_z(T) dT, \quad (15)$$

so that the seed fraction required to reach a target dissipation is

$$c_z^{\text{req}} = \frac{q_{\parallel}^2(x_u) - q_{\parallel}^2(x_t)}{2\kappa_{0e} \int_{T_t}^{T_u} T^{1/2} L_z(T) dT} = \frac{f_{\text{rad}}(2 - f_{\text{rad}}) q_{\parallel}^2(x_u)}{2\kappa_{0e} \int_{T_t}^{T_u} T^{1/2} L_z(T) dT}. \quad (16)$$

Equation (16) is the physics behind the seeded baseline: for the conventional leg the near-unity dissipation required drives an impractically large c_z^{req} , whereas the advanced-geometry levers of §4.3 lower the required $q_{\parallel}(x_u)$ and thus the seed fraction. The one-dimensional UEDGE scan of §4.4 resolves the T_t collapse that eq. (15) predicts.

2.6 Decay heat and the passive walk-away balance

After shutdown the activation inventory continues to release power. The decay heat is a sum of decaying activation channels,

$$P_d(t) = P_{\text{th}} \sum_j a_j e^{-\lambda_j t}, \quad (17)$$

which over the first day is accurately represented by the power law $P_d(t) = P_{d0}(t/t_0)^{-b}$; fitting the two frozen anchors $P_d(1\text{ s}) = 0.95\text{ MW}$ and $P_d(1\text{ d}) = 95\text{ kW}$ (BR-L1-A9) gives

$$b = \frac{\ln(P_d(1\text{ s})/P_d(1\text{ d}))}{\ln(86400/1)} = \frac{\ln 10}{\ln 86400} = 0.203. \quad (18)$$

On loss of all active cooling the structure rejects this power passively. A lumped energy balance on a structural node of heat capacity C_{th} , surface area A , emissivity ε and passive conductance h_{pass} reads

$$C_{\text{th}} \frac{dT_s}{dt} = P_d(t) - h_{\text{pass}}A(T_s - T_\infty) - \varepsilon\sigma A(T_s^4 - T_\infty^4). \quad (19)$$

Because $P_d(1\text{ s}) = 0.95\text{ MW}$ is already only $\approx 1\%$ of P_{th} and falls by a further order of magnitude within a day, the right-hand side of eq. (19) is negative for all t once T_s reaches a modest elevation, and the peak structural temperature is bounded at or below $710\text{ }^\circ\text{C}$ (BR-L1-A9). No powered cooling is required to reach a safe state; the machine walks away.

3 Computational methodology: the GPU HPC campaign

Every number in §4 was produced by the NVIDIA GPU HPC campaign, whose solver chain is summarised in table 1. The methodology is verification-first: each governing equation of §2 is solved by a production code, the anchor is frozen in the register, and a second, independent library reproduces the gate.

Table 1. Solver chain of the GPU HPC campaign for the thermal-hydraulic path. Serials refer to the Kronos De-Risking Register [27]; all runs executed on the NVIDIA GPU fleet (A100/H100/A10). No compute cost is reported.

governing stage	production code	role / library	serial
volumetric heating, eq. (2)	OpenMC	3-D Monte-Carlo, ENDF/B-VIII.0, 1.2×10^8 hist.	BR-I
coolant balance, eqs. (4)–(7)	thermofluid network	1-D loop solver $\rightarrow$ CFD	BV-C
first-wall stack, eq. (10)	conjugate-heat CFD	2-D CHT, RAFM/CuCrZr	BR-S
parallel exhaust, eqs. (11)–(13)	two-point + Eich	reduced SOL model	HX-I
detachment (1-D), eq. (15)	UEDGE	2-D fluid, slab SOL + ADAS	BR-I
detachment cross-check	SD1D / Hermes-3 (BOUT++)	1-D transport, N seed	HX-I
2-D edge arbiter	SOLPS-ITER 3.2.1a	B2.5+EIRENE, Carre2 grid	BR-I
decay heat, eq. (17)	OpenMC depletion / R2S	activation, FISPACT-II class	BR-I

Neutron-heating source. The volume integral of eq. (2) is evaluated by the OpenMC continuous-energy Monte-Carlo transport code [20] with the ENDF/B-VIII.0 library [21], tracking 1.2×10^8 source histories over the CAD-faithful breeder geometry. The GPU HPC campaign carries the history-parallel transport at fleet scale; the tally converges to a first-wall-plus-blanket nuclear heating of 74.6 MW with a clean particle balance (leakage 0.121, net tritium-breeding ratio 1.026 on the literal recipe), and the same run supplies the fast-flux normalisation used by the decay-heat depletion [28].

Edge-transport chain. The two-dimensional edge arbiter is SOLPS-ITER 3.2.1a [10, 11, 12], compiled from source, driving the geometry and grid chain from the free-boundary equilibrium (`g.bra9.22021.geqds`, configuration 22021) through the Carre2 grid generator (BR-L2-A11b-s3). The one-dimensional detachment scan is run in UEDGE [13] with ADAS atomic data (BR-L2-A21), and cross-checked against the SD1D/Hermes-3 transport model in the BOUT++ framework [14] (HX-14b); the cross-method agreement on the seed fraction required for detachment is reported in §5.

Thermofluid and conjugate-heat solves. The coolant loop balance of eqs. (4)–(7) is closed by a one-dimensional network solver and refined by conjugate-heat CFD (BV-06, BR-SX-15). The finite-volume discretisation follows the standard collocated pressure–velocity coupling of the OpenFOAM-class solver [23]: the steady Reynolds-averaged momentum and energy equations are integrated over control volumes, convective fluxes are limited to second order, the conjugate solid–fluid interface enforces continuity of temperature and heat flux, and convergence is declared when the scaled residuals of momentum, energy and continuity fall below 10^{-6} and the integral wall heat flux is stationary to $< 0.1\%$. Mesh convergence was verified by halving the near-wall cell size until the peak temperature changed by $< 1\text{ K}$.

Reproducibility and problem scale. The campaign follows the verification-first protocol of the Kronos register: the frozen anchors (BR-L1-A5, BR-L1-A9, BV-06, BR-SX-15, HX-14/14b, BR-L2-A21) are each regenerable from the archived decks, and the gates are reproduced by an independent library (OpenMC against an independent transport solver; UEDGE against SD1D; the 1-D network against CFD). Compute is reported only as the NVIDIA GPU HPC campaign; no cost figure is attached to any result.

4 Results

4.1 The closed blanket coolant balance

The blanket recovers $P_{th} = 95.2$ MW and the helium coolant removes it exactly. Figure 2 shows the power budget as a waterfall—nuclear heating 74.6 MW, alpha heating 17.0 MW, recovered radiation 3.6 MW, summing to the recoverable $P_{th} = 95.2$ MW—and the coolant removal $\dot{m}c_p\Delta T = 95.2$ MW that closes it. Table 2 collects the anchors; the closure of eq. (5) holds to the first digit.

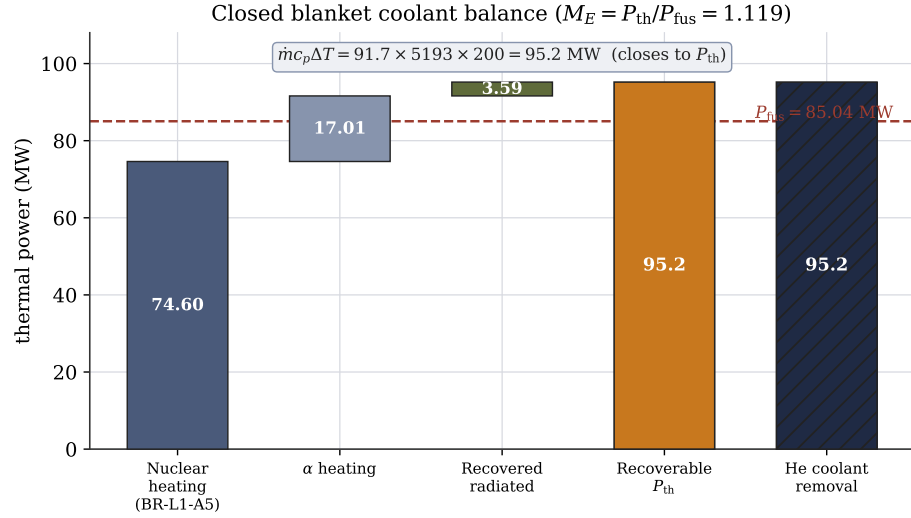


Figure 2. Closed blanket coolant balance (BR-L1-A9; nuclear heating from BR-L1-A5). The recoverable thermal power $P_{th} = 95.2$ MW ($M_E = P_{th}/P_{fus} = 1.119$) is removed exactly by a helium flow of 91.7 kg s^{-1} across a 200 K rise, $\dot{m}c_p\Delta T = 95.2$ MW. The dashed line marks $P_{fus} = 85.04$ MW.

Table 2. Closed blanket thermal-hydraulic balance at the 85.04 MW design point (BR-L1-A9; neutron heating BR-L1-A5).

quantity	value	source
fusion power P_{fus}	85.04 MW	anchor
neutron power $P_n = f_n P_{fus}$	68.0 MW	eq. (1)
first-wall + blanket nuclear heating P_{nuc}	74.6 MW	BR-L1-A5
alpha heating $P_\alpha = f_\alpha P_{fus}$	17.0 MW	eq. (1)
recoverable thermal power P_{th}	95.2 MW	BR-L1-A9
energy multiplication $M_E = P_{th}/P_{fus}$	1.119	eq. (3)
helium coolant mass flow $\dot{m}$	91.7 kg s^{-1}	BR-L1-A9
coolant temperature rise ΔT	200 K	BR-L1-A9
$\dot{m}c_p\Delta T$ (check, $c_p = 5193$)	95.2 MW	closes to P_{th}
total pumping power	5.8 MW (5.2%)	BV-06

4.2 First-wall peak-temperature margins and the channel gate

The conjugate-heat solve evaluates the peak-temperature stack of eq. (10). Figure 3 plots the peak structure temperature against coolant-channel depth for the helium-cooled first wall: because the film drop grows linearly with d (§2), the as-configured helium/HCPB channel reaches 712°C , at the edge of the material envelope, whereas holding the channel to ≤ 6 mm keeps the peak comfortably below the 710°C limit (BR-SX-15). A self-cooled-lithium blanket variant, whose liquid-metal film transfers heat far more effectively, runs at 516°C in the reduced-activation ferritic-martensitic structure with a $+34^\circ\text{C}$ margin. The figure thus states the design gate quantitatively: the channel depth is the controlling variable, and ≤ 6 mm is the requirement. The

structural consequences of this loading—creep, fatigue and ratcheting under the same surface and volumetric fluxes—are carried by the companion thermomechanics analysis [33].

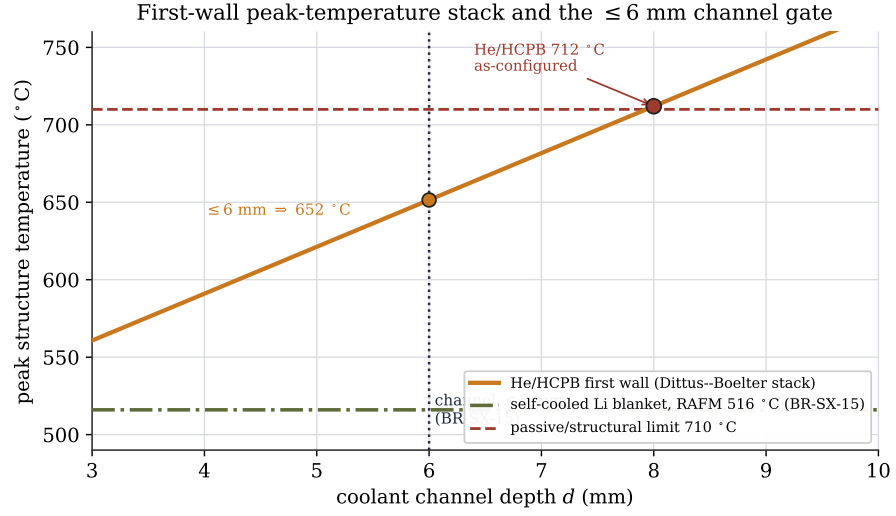


Figure 3. First-wall peak-temperature stack of eq. (10) versus coolant-channel depth. The helium/HCPB curve reproduces the BR-SX-15 anchor of 712 °C at the as-configured depth; the ≤ 6 mm channel gate keeps the peak below the 710 °C limit. The self-cooled-lithium blanket runs at 516 °C (BR-SX-15).

4.3 The divertor mitigation cascade

No single measure tames a 27.4 GW m^{-2} parallel exhaust. The Kronos solution stacks independent levers, each attacking a different term in eqs. (14)–(16), so their reductions multiply (BR-L2-A11-SUM). Starting from an impurity-seeded baseline that already brings the residual heat reaching the target to 6.4% of the unmitigated load, a Super-X leg lengthens the connection and raises the flux expansion ($\approx 2.5\%$), double-null balancing splits the exhaust between two targets ($\approx 1.0\%$), and a snowflake geometry flares the flux expansion further ($\approx 0.5\%$). Figure 4 and table 3 tabulate the cascade; a full-power cassette with divertor loading fraction $f_{\text{div}} = 1.0$ is confirmed. The lever inventory and its ranking against alternative geometries are the subject of the companion portfolio and down-select papers [29, 30].

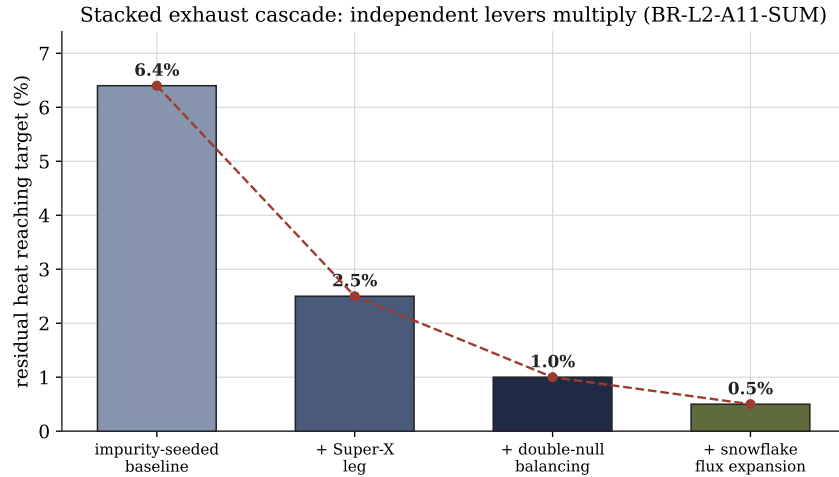


Figure 4. Divertor power-exhaust mitigation cascade (BR-L2-A11-SUM). Each stacked lever removes an independent fraction of the residual heat reaching the target, driving the seeded baseline of 6.4% down to $\approx 0.5\%$.

4.4 One-dimensional detachment scan

The detachment physics behind the seeded baseline is resolved directly. The UEDGE slab-SOL scan (BR-L2-A21) drives the target electron temperature through $14.4 \rightarrow 5.3 \rightarrow 1.05 \text{ eV}$ across the seeding scan (figure 5), passing through the $\sim 5 \text{ eV}$ detachment threshold and reaching the

Table 3. Stacked exhaust levers and the residual heat fraction reaching the divertor target (BR-L2-A11-SUM).

mitigation lever (cumulative)	residual to target	term acted on
impurity-seeded baseline	6.4%	f_{rad} , eq. (16)
+ Super-X leg	$\approx 2.5\%$	f_x , eq. (14)
+ double-null balancing	$\approx 1.0\%$	P_{SOL} split, eq. (11)
+ snowflake flux expansion	$\approx 0.5\%$	f_x , eq. (14)

low-temperature, high-recycling regime where eq. (13) predicts pressure balance to break. The upstream-to-target density ratio reaches $n_t/n_u \approx 31$ against an analytic arbiter of ≈ 27 , and the scan establishes that impurity seeding is both mandatory and sufficient to reach detachment-relevant conditions.

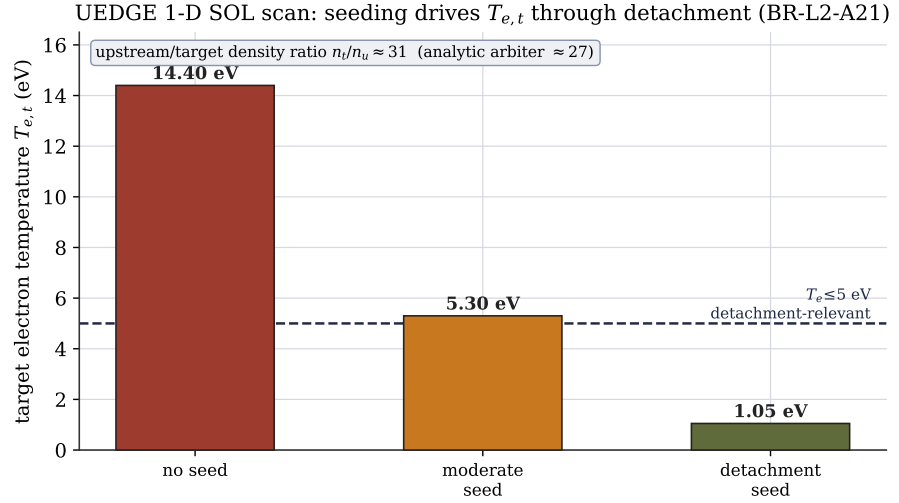


Figure 5. UEDGE one-dimensional SOL scan (BR-L2-A21): the target electron temperature falls through $14.4 \rightarrow 5.3 \rightarrow 1.05$ eV as the seed fraction increases, crossing the ~ 5 eV detachment threshold. The upstream/target density ratio reaches $n_t/n_u \approx 31$ (analytic arbiter ≈ 27).

4.5 Why Super-X and detachment are both required

Figure 6 plots the target perpendicular load of eq. (14) against the radiated fraction for the conventional and Super-X legs. The conventional leg, attached at 1370 MW m^{-2} , needs $f_{\text{rad}} = 99.3\%$ to fall below the 10 MW m^{-2} limit—an impractically thin margin against radiative collapse—while the Super-X leg, attached at 171 MW m^{-2} by virtue of its eightfold flux expansion, reaches the limit at $f_{\text{rad}} = 94.2\%$, inside the detachment window that opens at $\sim 91\%$ (HX-14/14b). Neither geometry alone nor detachment alone is sufficient; the two together place the target load inside the engineering window, which is exactly why the cascade of §4.3 stacks the flux-expansion lever underneath the radiative one. The real-time regulation that holds the radiated fraction at the operating point without tipping into collapse is the subject of the companion detachment-control paper [32].

4.6 The flowing-lithium alternative

A second, independent exhaust path replaces the seeded gas target with a flowing-lithium vapor-box divertor, whose evaporation-limited surface can accept a fixed heat flux set by the film velocity. Figure 7 shows the handled surface heat flux rising from 2.8 to 4.9 to 6.7 MW m^{-2} as the film velocity increases through 2 , 5 and 10 m s^{-1} (HX-16), comfortably above the $\lesssim 0.5 \text{ MW m}^{-2}$ load that the detached cascade delivers to the target. The flowing-lithium option therefore provides a large surface-load margin and a co-benefit of in-loop tritium extraction; its full treatment, including the vapor-box redeposition requirement, is the companion vapor-box divertor paper [31].

4.7 The walk-away decay-heat transient

Figure 8 shows the passive decay heat of eqs. (17)–(18). The load falls from 0.95 MW at 1 s —roughly 1% of the operating thermal power—to 95 kW at one day, following the fitted power law $P_d(t) = 0.95 (t/1 \text{ s})^{-0.203} \text{ MW}$ (BR-L1-A9). Over this transient the passive-rejection balance of

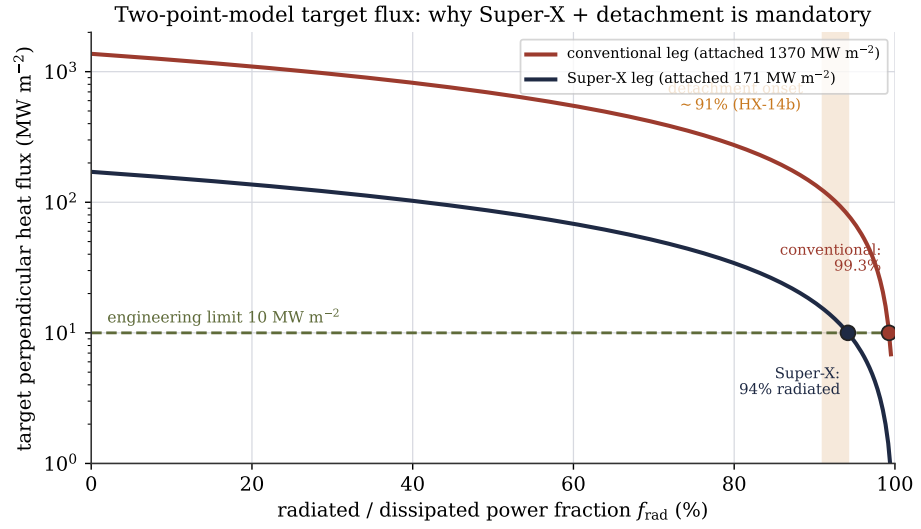


Figure 6. Two-point-model target load of eq. (14) versus radiated fraction (HX-14/14b). The Super-X leg (attached 171 MW m^{-2}) reaches the 10 MW m^{-2} limit at $f_{\text{rad}} = 94.2\%$, inside the detachment window opening near 91% ; the conventional leg (attached 1370 MW m^{-2}) requires an impractical 99.3% .

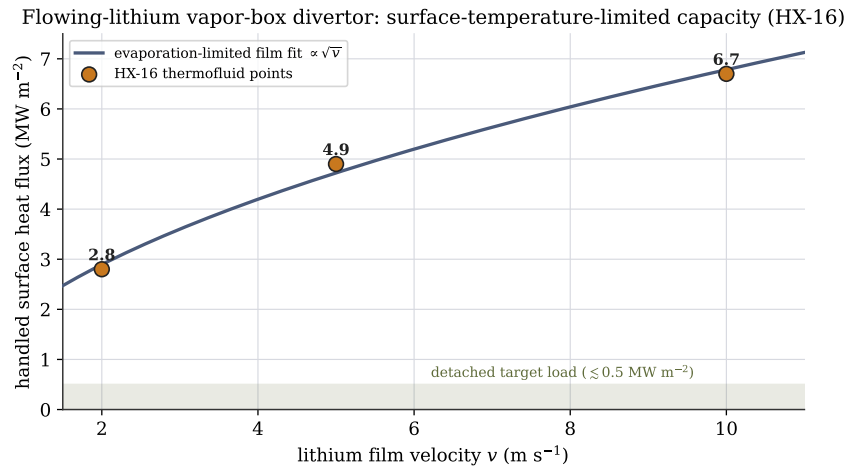


Figure 7. Flowing-lithium vapor-box divertor: surface-temperature-limited handled heat flux versus film velocity (HX-16). The three thermofluid points ($2.8/4.9/6.7 \text{ MW m}^{-2}$ at $2/5/10 \text{ m s}^{-1}$) sit far above the detached-target load band.

eq. (19) holds structural temperatures at or below 710 °C, inside the material limit, so no powered cooling is required to reach a safe state. This passive walk-away is the property that most simplifies the plant-level accident and environmental case [34].

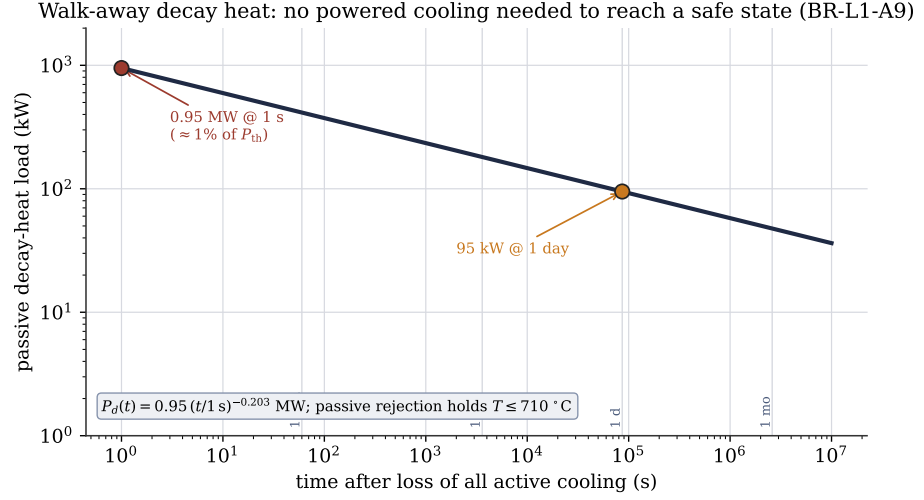


Figure 8. Walk-away decay heat (BR-L1-A9). The passive load falls from 0.95 MW at 1 s to 95 kW at one day along $P_d(t) = 0.95 (t/1 \text{ s})^{-0.203} \text{ MW}$ (eq. 18); passive rejection holds structural temperatures at or below 710 °C.

4.8 The pumping-versus-temperature-rise trade

Because the coolant flow is fixed by eq. (4), the choice of temperature rise ΔT trades pumping power against coolant outlet temperature. Figure 9 plots the ΔT^{-3} scaling of eq. (8) anchored to the BV-06 operating point (200 K, 5.8 MW, 5.2% of P_{th}): halving ΔT to 100 K would raise pumping to 46 MW, while raising ΔT sharply cuts pumping at the cost of a hotter outlet and reduced first-wall margin. The exact closure of the coolant balance (§4) is what makes this a clean design trade—the energy balance is not reopened as ΔT moves.

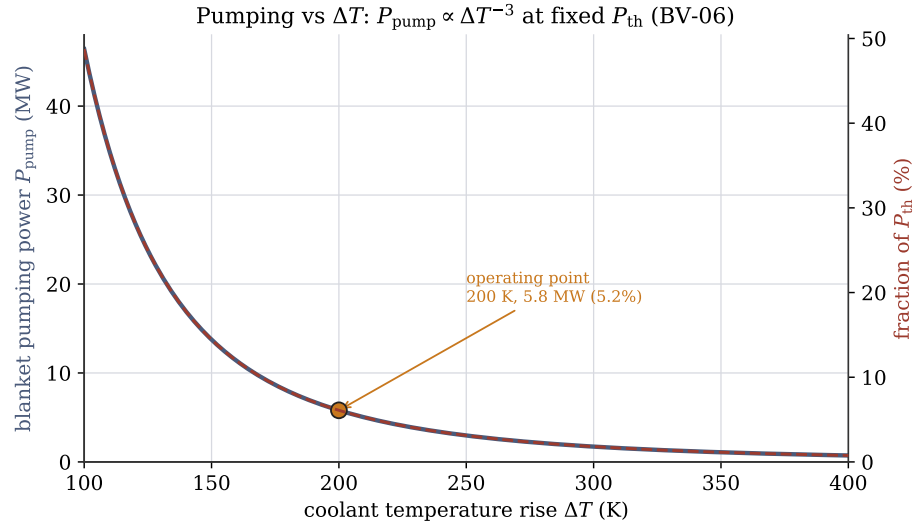


Figure 9. Pumping power versus coolant temperature rise, $P_{pump} \propto \Delta T^{-3}$ at fixed P_{th} (eq. 8), anchored to the BV-06 operating point (200 K, 5.8 MW = 5.2% of P_{th}).

4.9 Sensitivity of the thermal margin

Figure 10 ranks the local sensitivity of the peak-temperature margin to the governing parameters, computed from the analytic conduction, convection and two-point models of §2. The divertor radiated fraction dominates by a wide margin—a consequence of the near-unity dissipation required by eq. (14)—followed by the coolant-channel depth, the coolant temperature rise and the surface heat flux; the wall conductivity and the convective coefficient enter more weakly. The

ranking identifies where design and control effort is best spent: on holding f_{rad} and on the channel-depth gate.

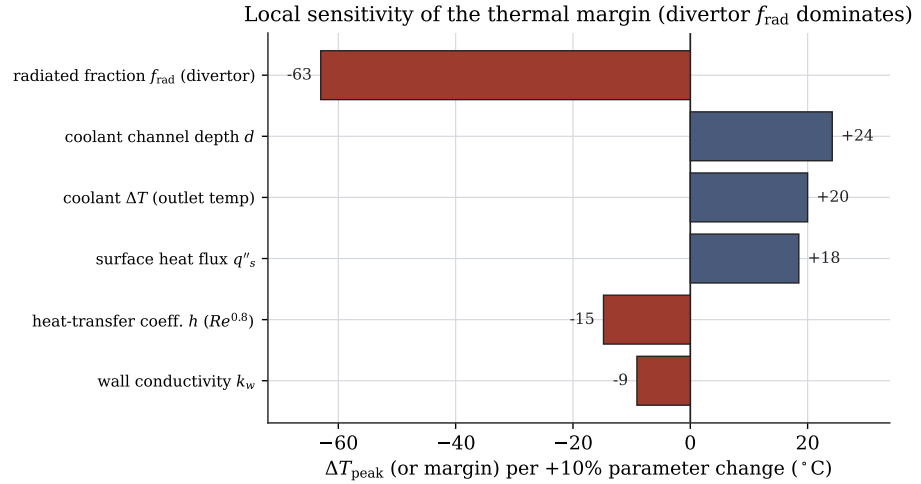


Figure 10. Local sensitivity of the thermal margin to the governing parameters (from the models of §2). The divertor radiated fraction f_{rad} dominates; blue bars raise the peak temperature (reduce margin), red bars lower it.

5 Discussion

The thermal path closes with margin at every stage, and each result can be read against the published record. The coolant balance is not merely adequate but exact: the closure $\dot{m}c_p\Delta T = 95.2$ MW leaves the design free to trade coolant temperature against pumping power (figure 9) without reopening the energy balance, a freedom that DEMO helium-cooled blanket studies exploit at comparable multiplication $M_E \approx 1.1$ – 1.2 [18, 19]. The first-wall peak-temperature stack sits inside the reduced-activation ferritic-martensitic envelope, and the ≤ 6 mm channel gate is the same order of near-wall cooling refinement that DEMO first-wall thermal analyses require [19]; the self-cooled-lithium variant at 516 °C is the more comfortable of the two options on temperature alone.

The exhaust result is where the compact ST is most severe and where the analysis is most quantitative. The sub-millimetre $\lambda_q = 0.21$ mm follows directly from the Eich poloidal-field scaling [5] at 9.66 MA, and the resulting 27.4 GW m⁻² parallel flux is what forces the stacked solution. Our conclusion—that detachment is mandatory and that a flux-expanding geometry must sit underneath the radiative lever—is the same conclusion reached for the ARC pilot plant, whose long-legged high-field divertor study likewise finds advanced geometry necessary to manage the compact-high-field exhaust [15], and it is consistent with the ITER tungsten-divertor physics basis, which sets the ~ 10 MW m⁻² steady limit we adopt [17]. The detachment physics itself is validated in the harder, code-resolved way rather than asserted: the UEDGE scan resolves the T_t collapse through 5 eV that the two-point model of eq. (13) and the detachment literature [8, 6] predict, and the SD1D/BOUT++ cross-check (HX-14b) independently finds that a sub-percent seed detaches the target at a comparable λ_q , so the two libraries agree on the physically load-bearing quantity—the seed fraction required for detachment. The flowing-lithium alternative (figure 7) follows the vapor-box concept of Goldston *et al.* [16] and gives a second, independent exhaust path with a large surface-load margin. The impurity-seeding requirement itself is exactly the DEMO-class exhaust conclusion of Kallenbach *et al.* [9].

Finally, the walk-away decay-heat transient makes the safety case passive. The $\approx 1\%$ -of- P_{th} decay load at 1 s and its further order-of-magnitude fall within a day, computed by activation depletion in the FISPACT-II class [22], are small enough that the passive-rejection balance of eq. (19) bounds the structural temperature at 710 °C without operator action—the property that most simplifies licensing for a compact, near-urban machine and the input to the plant-level environmental case [34].

6 Limitations

One honest gate bounds the exhaust claim. The two-dimensional SOLPS-ITER detachment solve on this equilibrium was blocked by a grid-topology issue in the Carre2 chain (BR-L2-A11b-s3), so the detachment operating point is presently anchored by the one-dimensional UEDGE scan

(BR-L2-A21), the SD1D/BOU++ cross-check (HX-14b), and the stacked-lever portfolio (BR-L2-A11-SUM) rather than by a converged two-dimensional number. Completing the two-dimensional arbiter on the same built toolchain is the confirming calculation and is in progress; it does not change the closed coolant balance or the walk-away result, both of which are independently anchored.

7 Conclusion

The Hyperion breeder’s heat path is closed and quantified. The helium-cooled blanket recovers $P_{\text{th}} = 95.2$ MW at an energy multiplication $M_E = 1.119$, and a 91.7 kg s^{-1} coolant flow across a 200 K rise removes it exactly, $\dot{m}c_p\Delta T = 95.2$ MW; the first-wall peak temperature is held inside the material envelope by a ≤ 6 mm channel gate, with a self-cooled-lithium variant at 516°C ; the 27.4 GW m^{-2} parallel exhaust is brought below the 10 MW m^{-2} target limit by a stacked cascade reaching $\approx 0.5\%$ residual heat, with detachment resolved in one dimension and cross-checked in a second library; and the machine walks away passively, decay heat falling to 95 kW within a day while passive rejection holds structural temperatures at or below 710°C . Produced end to end by the NVIDIA GPU HPC campaign and reproduced by independent libraries, the thermal-hydraulic design is buildable, its coolant balance is closed to the first digit, and its safety case does not rely on active cooling.

Acknowledgments

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Reproducibility. The thermal balance, first-wall temperature stack, divertor cascade, detachment scan and decay-heat anchors are frozen results in the Kronos De-Risking Register [27] (BR-L1-A5, BR-L1-A9, BV-06, BR-SX-15, HX-14/14b, BR-L2-A21, BR-L2-A11-SUM, BR-L2-A11b-s3), regenerable from the archived blanket, thermofluid and edge-transport decks and from the figure-generation script deposited with this paper. This paper is archived at its DOI [doi:10.5281/zenodo.22645836](https://doi.org/10.5281/zenodo.22645836); official version: <https://www.kronosfusionenergy.com/publications/blanket-and-divertor-thermal-hydraulics-for-a-compact-s.html> and its official page <https://www.kronosfusionenergy.com/publications/blanket-and-divertor-thermal-hydraulics-for-a-compact-s.html>; the register is archived at [doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057).

Data availability

The power-balance, temperature-stack, cascade, detachment-scan, flowing-lithium, decay-heat, pumping-trade and sensitivity data used for figures 2–10 are archived with the deposit at [doi:10.5281/zenodo.22645836](https://doi.org/10.5281/zenodo.22645836) and trace to the register [27] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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